What Does the Global Latent Workspace Learn?

Anonymous Author(s)

Abstract

Cognition requires more than parallel processing of separate sensory streams. It demands shared representations that can support integration and flexible reuse across perception, memory, and action. Global Workspace Theory offers a framework of how this is achieved: specialized processors operate in parallel while a shared workspace broadcasts selected information across brain areas. Recent deep learning architectures make this idea concrete by connecting modality-specific modules through a common latent space. Yet it remains unclear what structure these training procedures actually produce and when the resulting workspace is genuinely multimodal rather than merely compressive. We develop a theoretical framework in a simplified linear setting where the problem admits closed-form solutions. We show that the structure of the learned workspace depends critically on what it is trained to do. If each module only reconstructs its own input, the workspace compresses each modality independently. If modules translate into each other's format the workspace aligns with the statistical structure common to both modalities. This suggests that a workspace becomes integrative only when training pressures translation across modalities, not compression within each modality alone.

Introduction

Multimodal integration is fundamental to cognition because signals in one modality often constrain the interpretation of signals in another—as when visual lip movements disambiguate an auditory speech signal under noisy conditions (McGurk & MacDonald, 1976). Such cases suggest that cognition depends not only on modality-specific processing streams, but also on mechanisms that integrate information across modalities into a coherent and usable representation. Neuroscientific evidence supports this view: regions such as the superior temporal sulcus and middle temporal gyrus are consistently implicated in audiovisual integration (Beauchamp, 2005; Gao et al., 2023; Okada et al., 2013; Watson et al., 2014), suggesting a distributed cortical network rather than a single integration site. However, identifying multisensory regions does not by itself explain how information is stabilized, coordinated, and made available for flexible cognitive use. The Global Workspace framework (Baars, 1993; Dehaene et al., 1998) addresses this gap by proposing that specialized processors operate in parallel, while only a limited subset of information gains access to a shared workspace from which it can be broadcast to systems supporting memory, report, and action (Dehaene & Changeux, 2011; Dehaene et al., 1998). Recent deep learning architectures make this idea concrete by linking modality-specific encoders through a shared latent space trained with cycle-consistency objectives—requiring that information passed from one modality to another and back is faithfully preserved (Devillers et al., 2024; VanRullen & Kanai, 2021). Yet building such a system does not, by itself, explain what makes it work. What representational structure must a shared workspace have in order to support genuine multimodal integration rather than mere compression? We address this question theoretically. In a linear two-modality setting, we analyze the global workspace that emerges under different cycle-consistency objectives and show that the decoder determines what the workspace learns: self-reconstruction yields independent principal component analysis (PCA) solutions that preserve modality-specific variance, whereas cross-modal translation recovers the subspaces of canonical correlation analysis (CCA), organizing the workspace around shared statistical structure (Figure 1 A).

Theoretical derivation

Let $x \in \mathbb{R}^{d_x}$ and $y \in \mathbb{R}^{d_y}$ denote zero-mean random vectors corresponding to two modalities, with covariance matrices $\Sigma_{xx} = \mathbb{E}[xx^\top]$, $\Sigma_{yy} = \mathbb{E}[yy^\top]$, and cross-covariance $\Sigma_{xy} = \mathbb{E}[xy^\top]$. We consider k-dimensional linear encoders $z_x = W_x x$ and $z_y = W_y y$, where $W_x \in \mathbb{R}^{k\times d_x}$ and $W_y \in \mathbb{R}^{k\times d_y}$, together with linear decoders $\hat{y} = V_y z_x$ and $\hat{x} = V_x z_y$, where $V_y \in \mathbb{R}^{d_y\times k}$ and $V_x \in \mathbb{R}^{d_x\times k}$. The full cycle-consistency loss:

$$\mathcal{L}_{\text{cy}} = \mathbb{E}\left[\|x - V_x W_y V_y W_x x\|^2\right]_x + \mathbb{E}\left[\|y - V_y W_x V_x W_y y\|^2\right]_y .$$

Throughout, we impose whitening constraints $W_x \Sigma_{xx} W_x^\top = I_k$ and $W_y \Sigma_{yy} W_y^\top = I_k$, which remove trivial scale degrees of freedom and ensure the latent dimensions are decorrelated. We also define $U := W_x \Sigma_{xx}^{1/2}$ and $V := W_y \Sigma_{yy}^{1/2}$, so that $UU^\top = I_k$ and $VV^\top = I_k$. How does the choice of decoder shape the workspace? We consider two cases: decoders optimized to reconstruct each modality from its own encoding (self-reconstruction), and decoders optimized to translate between modalities (cross-translation). The two choices lead to qualitatively different solutions. **Self-reconstruction yields PCA.** If the decoders are optimized to reconstruct each modality

In submission to the *9th Conference on Cognitive Computational Neuroscience* (CCN 2026).

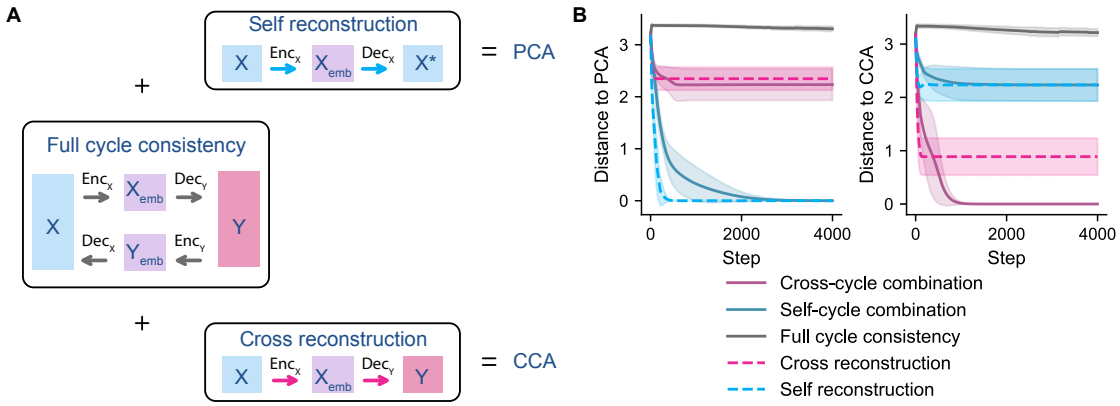

Figure 1: **(A)** Training objectives for one cycle (the symmetric Y cycle omitted). Full cycle consistency maps each modality through the other and back. Combined with self-reconstruction decoders (top), it reduces to PCA; combined with cross-reconstruction decoders (bottom), it recovers CCA. **(B)** Linear-Gaussian validation. Subspace distance to PCA (left) and CCA (right) targets during optimization for five objectives. Solid: cycle-consistency variants; dashed: single-step baselines. Shading: ± 1 SD across five seeds.

from their own latent embedding, the cycle-consistency loss simply preserves the directions of greatest variance within each input.

Proposition 1. *Suppose V_x and V_y are chosen as the optimal linear self-reconstruction decoders, i.e., Let $V_x \in \arg\min_{\tilde{V}_x} \mathbb{E}\|x - \tilde{V}_x W_x x\|^2$ and $V_y \in \arg\min_{\tilde{V}_y} \mathbb{E}\|y - \tilde{V}_y W_y y\|^2$. Then $\arg\min \mathcal{L}_{\text{cy}}$ is achieved when the row spaces of U and V are the top-k eigenspaces of Σ_{xx} and Σ_{yy}.*

Cross-translation yields CCA. If the decoders are instead optimized to translate each modality into the other's format, the cycle-consistency loss selects the directions along which the two modalities are most mutually predictive.

Proposition 2. *Let $V_y \in \arg\min_{\tilde{V}_y} \mathbb{E}\|y - \tilde{V}_y W_x x\|^2$ and $V_x \in \arg\min_{\tilde{V}_x} \mathbb{E}\|x - \tilde{V}_x W_y y\|^2$. Then $\arg\min \mathcal{L}_{\text{cy}}$ is achieved when the row spaces of U and V are the top-k left and right singular subspaces of $\Sigma_{xx}^{-1/2}\Sigma_{xy}\Sigma_{yy}^{-1/2}$.*

Empirical validation

We validated these predictions in a controlled linear-Gaussian setting by tracking the projection Frobenius distance between learned encoder subspaces and the theoretical PCA and CCA targets across optimization (Figure 1 B). As predicted, self reconstruction and self-cycle combination converge to PCA but not CCA, while the cross-cycle combination converges to CCA but not PCA. Cross reconstruction alone moves toward CCA but plateaus before recovering it. Full cycle consistency alone, converges to neither target cleanly.

Conclusion

Our analysis identifies a sharp condition under which a global workspace supports genuine multimodal integration. Cycle consistency alone is not sufficient: when paired with self-reconstruction, it preserves modality-specific variance and yields independent PCA solutions. Integration emerges only when the workspace is shaped by pressure to translate across modalities, in which case it recovers the canonical subspaces that capture shared statistical structure. This distinction mirrors a pattern across existing vision–language architectures. Models trained with cross-modal objectives (e.g., contrastive alignment in CLIP (Radford et al., 2021)) learn shared representations that transfer broadly across tasks. Models trained only to reconstruct within each modality do not, even when they share an architectural bottleneck (e.g., deep generative models (Suzuki et al., 2016)). Our results offer a theoretical account of why: cross-modal pressure selectively retains what is mutually informative across modalities, whereas reconstruction preserves modality-specific variance that may be irrelevant to integration. For Global Workspace theories more broadly, this means that the capacity to broadcast useful information across specialized systems is not a passive property of a shared bottleneck but depends on the learning objective that shapes it.

Acknowledgments

We used Claude to assist with editing purpose and debugging experiment code.

References

Baars, B. J. (1993). *A cognitive theory of consciousness*. Cambridge University Press.

Beauchamp, M. S. (2005). See me, hear me, touch me: Multisensory integration in lateral occipital-temporal cortex. *Current opinion in neurobiology*, *15*(2), 145–153.

Dehaene, S., & Changeux, J.-P. (2011). Experimental and theoretical approaches to conscious processing. *Neuron*, *70*(2), 200–227.

Dehaene, S., Kerszberg, M., & Changeux, J.-P. (1998). A neuronal model of a global workspace in effortful cognitive tasks. *Proceedings of the national Academy of Sciences*, *95*(24), 14529–14534.

Devillers, B., Maytié, L., & VanRullen, R. (2024). Semi-supervised multimodal representation learning through a global workspace. *IEEE Transactions on Neural Networks and Learning Systems*, *36*(5), 7843–7857.

Gao, C., Green, J. J., Yang, X., Oh, S., Kim, J., & Shinkareva, S. V. (2023). Audiovisual integration in the human brain: A coordinate-based meta-analysis. *Cerebral Cortex*, *33*(9), 5574–5584.

McGurk, H., & MacDonald, J. (1976). Hearing lips and seeing voices. *Nature*, *264*(5588), 746–748.

Okada, K., Venezia, J. H., Matchin, W., Saberi, K., & Hickok, G. (2013). An fmri study of audiovisual speech perception reveals multisensory interactions in auditory cortex. *PloS one*, *8*(6), e68959.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. (2021). Learning transferable visual models from natural language supervision. *International Conference on Machine Learning*, 8748–8763.

Suzuki, M., Nakayama, K., & Matsuo, Y. (2016). Joint multimodal learning with deep generative models. *arXiv preprint arXiv:1611.01891*.

VanRullen, R., & Kanai, R. (2021). Deep learning and the global workspace theory. *Trends in Neurosciences*, *44*(9), 692–704.

Watson, R., Latinus, M., Charest, I., Crabbe, F., & Belin, P. (2014). People-selectivity, audiovisual integration and heteromodality in the superior temporal sulcus. *cortex*, *50*, 125–136.